SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE:CSA51   COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO* (BL3, BL6)

Assessment Tool Weightage: 20%

QUESTIONS: 5   TOTAL MARKS: 25

Annexure A

Industry Problem-Solving Task

Code of Conduct:

I Atchaya Chandran R(192521394) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

R . Atchay

Signature of the student:

Annexure A

1. SSL/TLS Overhead and Web Server Performance OptimizationA web server uses TLS 1.3 for secure communication. Each HTTP response carries 2000 bytes of payload. TLS adds 25 bytes (record header + authentication tag) and handshake overhead of 3 KB per session. The server handles 10,000 requests per second, with 1,000 new TLS handshakes per second.

Task:A. Calculate the total data transmitted per second including TLS overhead.B. Determine the percentage overhead due to TLS encryption.C. Estimate bandwidth consumed due to handshake overhead per second.D. Analyze the trade-off between security and latency, and propose optimization techniques (e.g., session resumption, HTTP/2, or QUIC).

2. PGP-Based Email Encryption Load in Enterprise SystemAn enterprise email server encrypts emails using PGP. Each email contains a 10 MB attachment. PGP introduces 800 bytes overhead for key encryption and signatures. The system processes 5,000 emails per day.

Task:A. Calculate the total data transmitted per day including overhead.B. Determine the percentage increase in storage due to PGP.C. Analyze the computational overhead during encryption/decryption.D. Recommend optimization strategies (e.g., compression, batching, or hardware acceleration).

3. IPSec Transport vs Tunnel Mode Performance ComparisonA company compares IPSec transport mode and tunnel mode for securing internal communications. Original packet size = 1000 bytes payload + 20-byte header.Transport mode adds 30 bytes, while tunnel mode adds 50 bytes overhead. Traffic volume = 20,000 packets per second.

Task:A. Compute total packet size for both modes.B. Calculate bandwidth usage for each mode per second.C. Compare efficiency and security benefits of both modes.D. Recommend the appropriate mode for internal vs external communication with justification.

4. S/MIME Certificate Management in Large OrganizationAn organization manages 12,000 employees using S/MIME certificates. Each certificate is 2 KB, and certificate renewal happens every year. During renewal, both old and new certificates are stored simultaneously for 1 month.

Task:A. Calculate total storage required during the renewal period.B. Estimate certificate distribution overhead across the network.C. Analyze challenges in certificate lifecycle management at scale.D. Propose optimization strategies (e.g., automation, centralized PKI, or short-lived certificates).

5. Email Encryption Latency in Cloud-Based SystemsA cloud email service encrypts outgoing emails using AES (data) and RSA (key exchange). AES encryption takes 0.2 ms per MB, and RSA key exchange takes 5 ms per email. Each email has 8 MB attachment. The system processes 8,000 emails per hour.

Task:A. Calculate total encryption time per email.B. Estimate total processing time per hour.C. Analyze the performance bottleneck between symmetric and asymmetric operations.D. Recommend optimization strategies (e.g., ECC adoption, parallel processing, or hardware acceleration).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

1. SSL/TLS Overhead and Web Server Performance Optimization

Given Data:

- Payload per HTTP response =
- TLS record overhead per request =
- Handshake overhead per session =
- Total HTTP requests per second =
- New TLS handshakes per second =

Task Solutions:

A. Total Data Transmitted per Second

- Payload data per second:

- TLS Record overhead per second:

- TLS Handshake overhead per second:

- Total Data Transmitted:

B. Percentage Overhead Due to TLS Encryption

- Total Overhead:

- Percentage Overhead:

C. Bandwidth Consumed Due to Handshake Overhead per Second

- Handshake Bandwidth:

D. Trade-off Analysis & Optimization Techniques

- Trade-off: TLS 1.3 provides strong confidentiality and integrity, but frequent handshakes add latency (1-RTT) and consume significant CPU/bandwidth.
- Optimization Techniques:
  1. TLS Session Resumption (Session Tickets / 0-RTT): Eliminates full handshake latency and reduces overhead for returning clients.
  2. HTTP/2 or HTTP/3 (QUIC): Multiplexes multiple requests over a single TLS connection, drastically reducing full handshakes.
  3. TLS Offloading / Hardware Acceleration: Offloads cryptography processing to dedicated hardware (e.g., HSMs or smartNICs).

2. PGP-Based Email Encryption Load in Enterprise System

Given Data:

- Attachment size per email =
- PGP overhead per email =
- Total emails processed per day =

Task Solutions:

A. Total Data Transmitted per Day

- Size per encrypted email:

- Total Data Transmitted per day:

B. Percentage Increase in Storage Due to PGP

- Percentage Overhead:

C. Analysis of Computational Overhead

- PGP uses hybrid encryption: asymmetric encryption (RSA/ECC) for key exchange and signature, and symmetric encryption (AES) for the payload.
- Symmetric Encryption (AES): CPU-intensive due to bulk data encryption/decryption.
- Asymmetric Operations: High mathematical computational complexity per message for digital signing and session key wrapping.

D. Optimization Strategies

1. Integrated Compression (ZIP/ZLIB): Compress attachments before encryption to offset the encryption metadata overhead.
2. Hardware Acceleration (AES-NI): Use CPU instruction sets dedicated to symmetric encryption.
3. Asymmetric Key Upgrade: Migrate from RSA to Elliptic Curve Cryptography (ECC) to speed up key negotiation and digital signatures.

3. IPSec Transport vs Tunnel Mode Performance Comparison

Given Data:

- Original packet size =
- Transport mode overhead =
- Tunnel mode overhead =
- Traffic volume =

Task Solutions:

A. Compute Total Packet Size

- Transport Mode Packet Size:

- Tunnel Mode Packet Size:

B. Calculate Bandwidth Usage per Second

- Transport Mode Bandwidth:

- Tunnel Mode Bandwidth:

C. Efficiency and Security Comparison

Metric / Aspect	Transport Mode	Tunnel Mode
Header Overhead	(Lower)	(Higher)
Bandwidth Usage	(More Efficient)	
Payload Protection	Encrypts payload only	Encrypts entire IP packet (Payload + Inner Header)
Identity Hiding	Source/Destination IP exposed	Hides internal IP addresses

D. Mode Recommendations & Justification

- Internal Communication: Recommend Transport Mode. End-to-end communication between internal hosts doesn't require IP header encapsulation, saving bandwidth and processing power.
- External Communication: Recommend Tunnel Mode. Essential for Gateway-to-Gateway (site-to-site VPN) or Gateway-to-Host scenarios to hide internal network architecture and ensure gateway security.

4. S/MIME Certificate Management in Large Organization

Given Data:

- Total employees =
- Certificate size =
- Overlap during annual renewal = (Both old and new active)

Task Solutions:

A. Total Storage Required During Renewal Period

During the renewal period, each user holds certificates (1 old + 1 new):

B. Certificate Distribution Overhead Across Network

- Distributing new certificates to all users:

- Including Certificate Revocation Lists (CRLs) or OCSP responses increases overall bandwidth overhead linearly with user count.

C. Lifecycle Management Challenges at Scale

1. Key Rollover & Decryption: Need to safely archive expired private keys so employees can decrypt historical encrypted emails.
2. Certificate Expiration Outages: Manual renewals often lead to missed expirations, causing broken email workflows.
3. Distribution Delays: Propagating public keys and updated CRL/OCSP across multi-device client environments causes trust issues.

D. Optimization Strategies

1. Automated PKI Enrollment (EST / SCEP / ACME): Automates key generation, issuance, and renewal without human intervention.
2. Centralized Key Management System (KMS): Manages key archiving to seamlessly decrypt historical messages.
3. Lightweight Directory Access Protocol (LDAP) / Active Directory: Centralizes public key distribution across the enterprise directory.

5. Email Encryption Latency in Cloud-Based Systems

Given Data:

- AES encryption rate =
- RSA key exchange time =
- Attachment size =
- System throughput =

Task Solutions:

A. Total Encryption Time per Email

- AES Symmetric Encryption Time:

- RSA Key Exchange Time:

- Total Time per Email:

B. Total Processing Time per Hour

- Total Processing Time:

C. Performance Bottleneck Analysis

- Asymmetric operation (RSA) takes (of total time), whereas symmetric encryption (AES) for an entire payload takes only ().
- Conclusion: The primary performance bottleneck is the asymmetric key exchange operation (RSA), despite the payload being relatively large.

D. Optimization Strategies

1. Transition to ECC (ECDH / ECDSA): Elliptic Curve Cryptography requires significantly smaller key sizes and offers much faster key exchanges than RSA.
2. Asynchronous / Parallel Processing: Process RSA key exchanges concurrently across multi-core systems or offload them to dedicated Cloud HSMs.
3. AES-NI Hardware Acceleration: Accelerate AES payload operations directly at the CPU level.